Generating Generalised Ground-State Ansätze from Few-Body Examples

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The Problem

$$|\psi\rangle = \sum_{\sigma_1,\ldots,\sigma_N} \psi_{\sigma_1,\ldots,\sigma_N} |\sigma_1,\ldots,\sigma_N\rangle$$

$$\downarrow$$

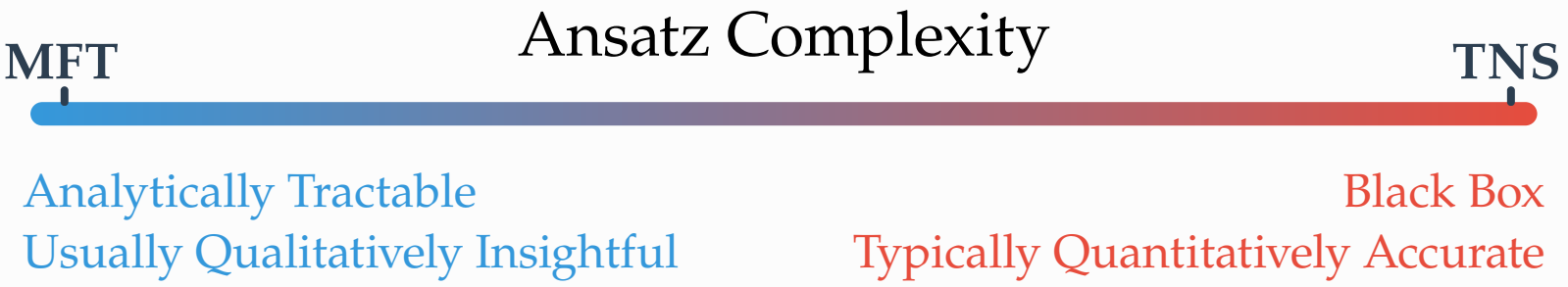
$$O(d^N)$$

$$\text{Physical System} \implies \text{Redundancies}$$

Goal: Find *effective compressed description

$$\underbrace{\qquad\qquad\qquad\qquad\qquad\qquad}_{\text{ansatz}}$$

Example: Describing Ground States



Example: Describing Ground States



Analytically Tractable
Usually Qualitatively Insightful

Black Box
Typically Quantitatively Accurate

$$H = \sum_j \vec{S}_j \cdot \vec{S}_{j+1} + \frac{1}{3}(\vec{S}_j \cdot \vec{S}_{j+1})^2, \quad |\psi_g\rangle = \sum_{\sigma_1,\ldots,\sigma_N} \mathrm{Tr}(A^{\sigma_1} A^{\sigma_2} \cdots A^{\sigma_N}) \, |\sigma_1, \ldots, \sigma_N\rangle$$

$$A^+ = \sqrt{\frac{2}{3}} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$A^0 = \sqrt{\frac{1}{3}} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$A^- = \sqrt{\frac{2}{3}} \begin{bmatrix} 0 & 0 \\ -1 & 0 \end{bmatrix}$$

Gauge Invariance

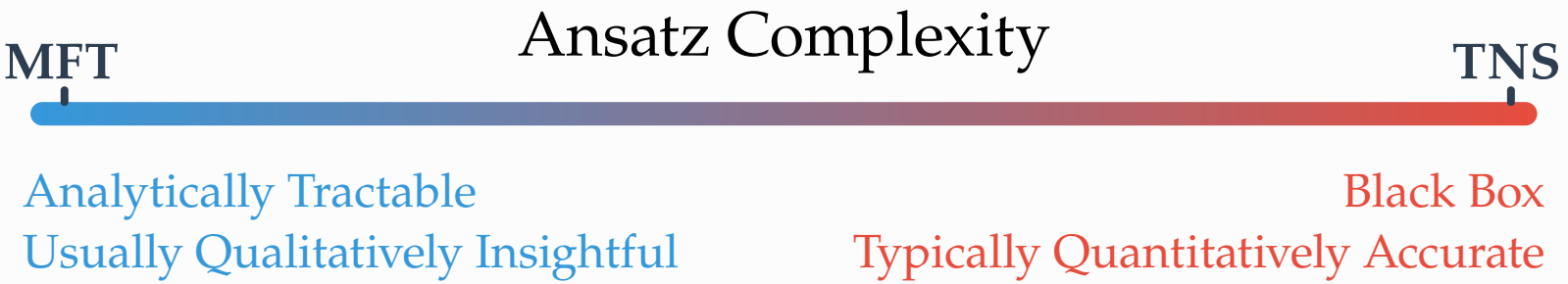
$$\tilde{A}^+ = \begin{bmatrix} -1.96 + 0.05i & -0.67 - 1.81i \\ 0.61 - 1.90i & 1.96 - 0.05i \end{bmatrix}$$

$$\tilde{A}^0 = \begin{bmatrix} 0.07 + 0.01i & -1.51 + 2.46i \\ 2.73 - 0.90i & -0.07 - 0.01i \end{bmatrix}$$

$$\tilde{A}^- = \begin{bmatrix} -0.54 + 2.04i & 2.14 - 0.24i \\ 1.68 + 1.22i & 0.54 - 2.04i \end{bmatrix}$$

Affleck, Kennedy, Lieb, and Tasaki (AKLT)

Example: Describing Ground States



Goal:

Minimise Complexity

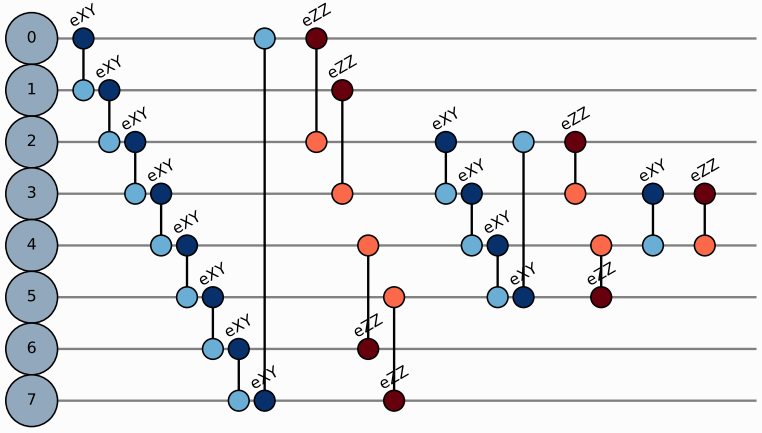
←

Preserve Accuracy

Ansatz "Complexity"?

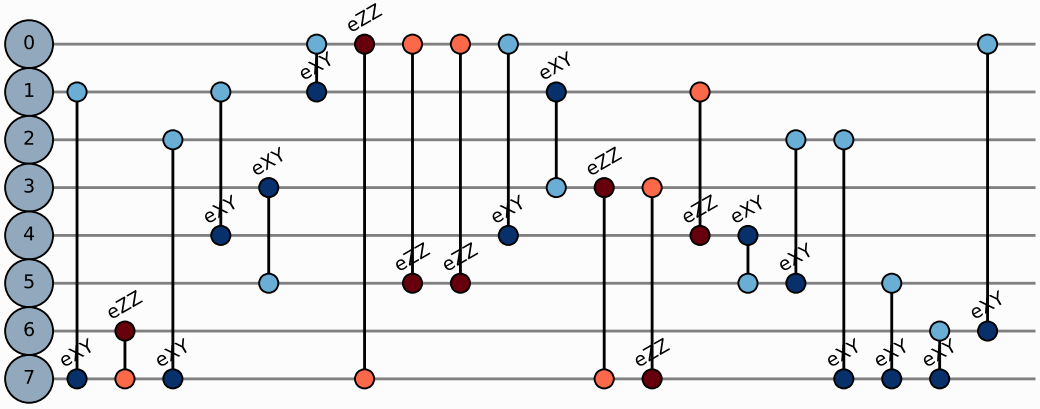
`01010101010101010101`

```
print("01" * 10)
```



`10000000011011110101`

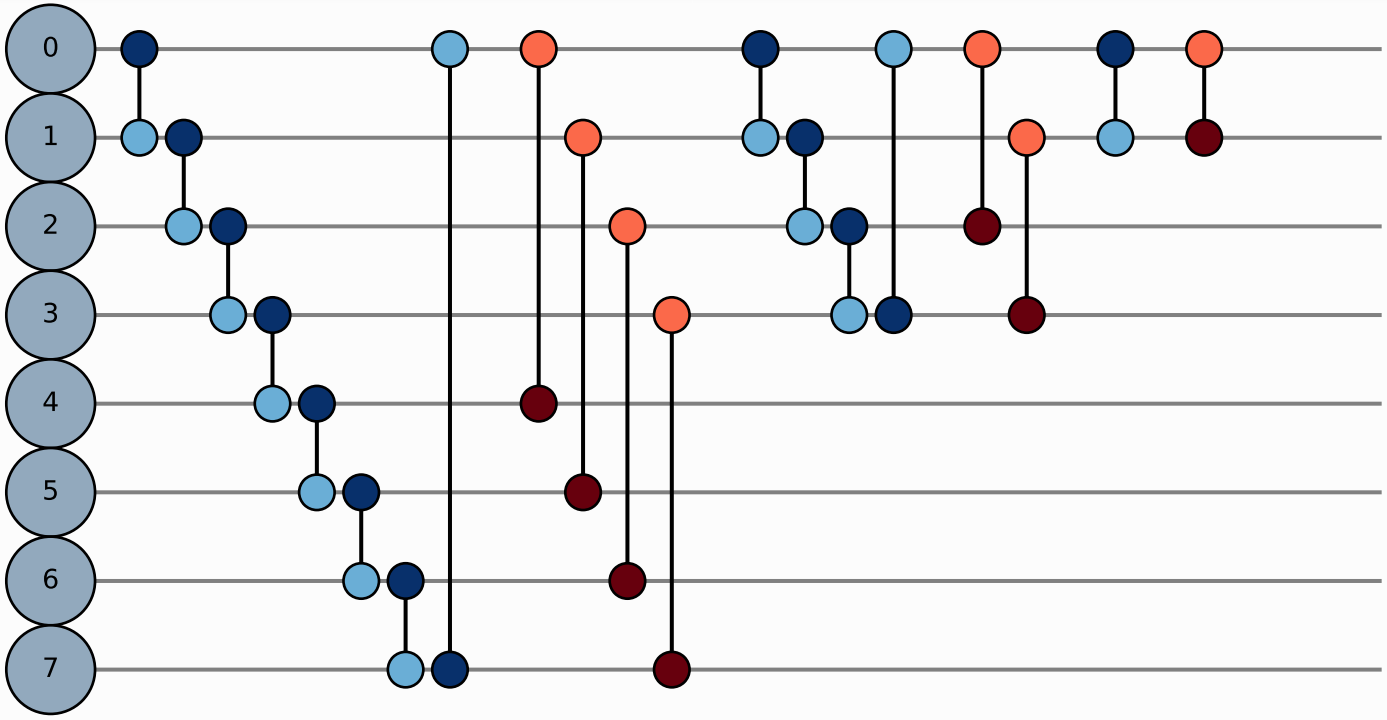
```
print("10000000011011110101")
```



Algorithmic Description

Compressed *Algorithmic* Descriptions

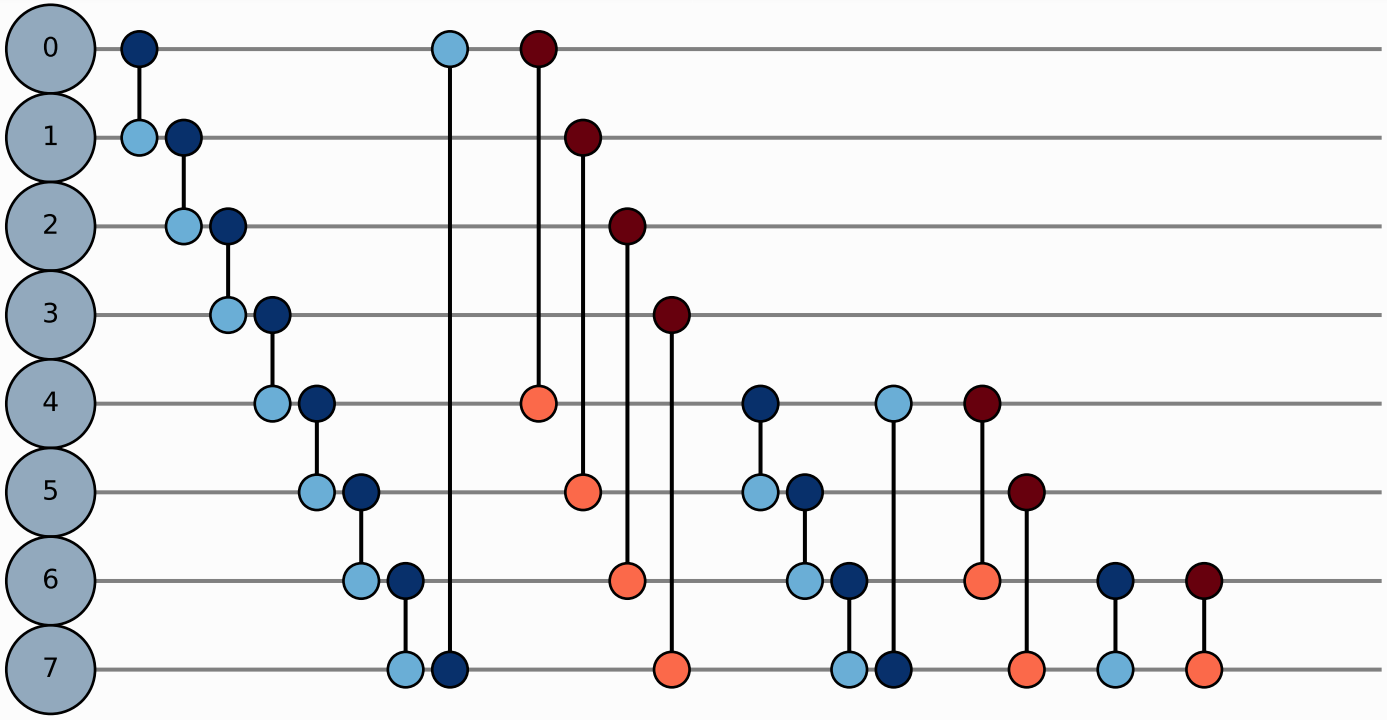
```
1 cycle = Qcycle(step=1, stride=1, k=2, boundary=0)
2 mask  = Qmask(bp="*!", k=2)
3 mask  = Qmask(bp="!*", k=2)
4 mask  = Qmask(bp="!*!",k=2)
5 mask  = Qmask(bp="01", k=2)
6 Qinit(8) + (cycle + mask)*3
```



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Compressed *Algorithmic* Descriptions

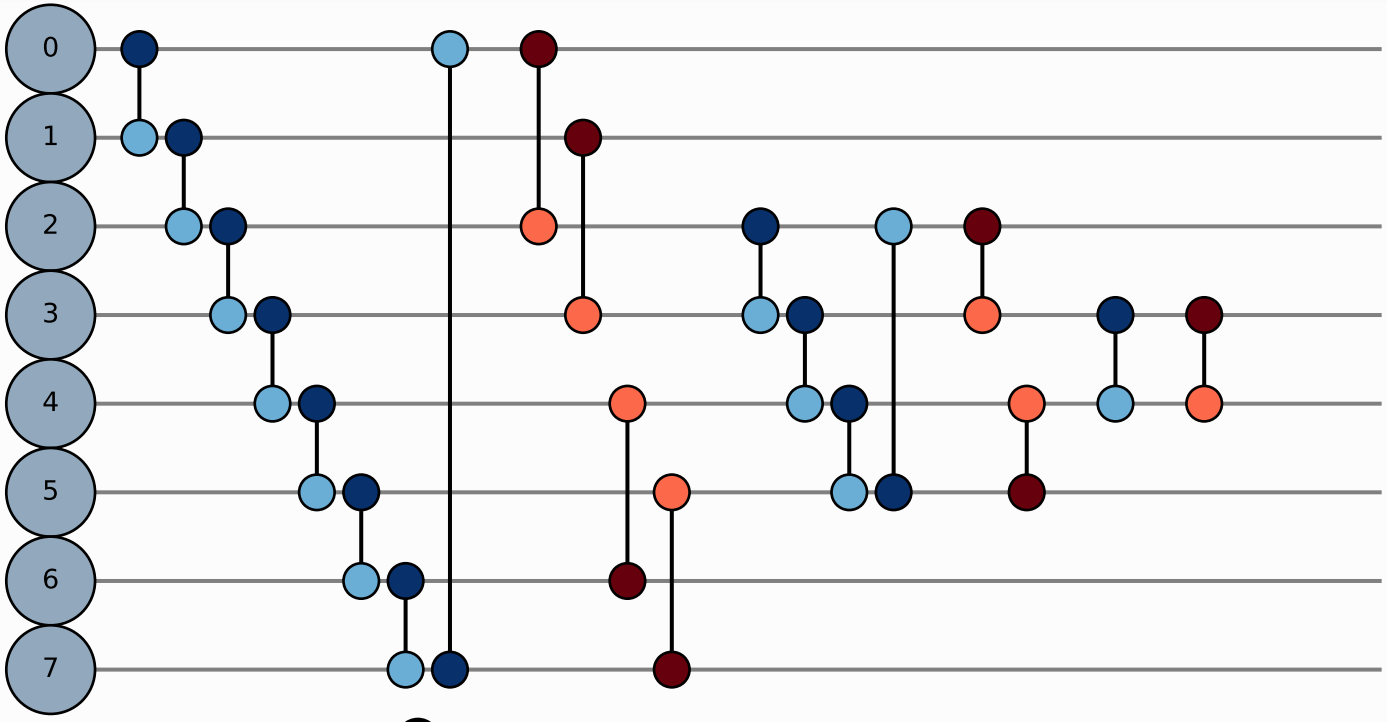
```
1 cycle = Qcycle(step=1, stride=1, k=2, boundary=0)
2 mask  = Qmask(bp="*!", k=2)
3 mask  = Qmask(bp="!*", k=2)
4 mask  = Qmask(bp="!*!",k=2)
5 mask  = Qmask(bp="01", k=2)
6 Qinit(8) + (cycle + mask)*3
```



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Compressed *Algorithmic* Descriptions

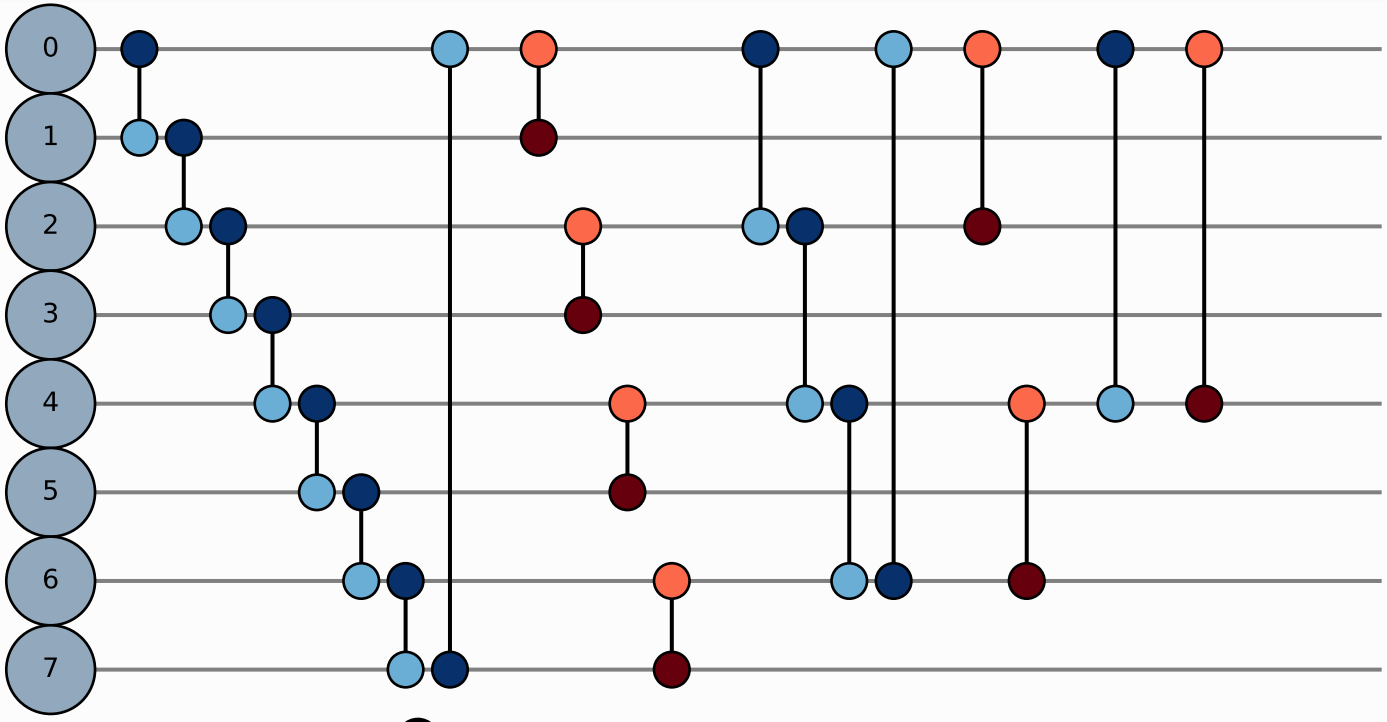
```
1 cycle = Qcycle(step=1, stride=1, k=2, boundary=0)
2 mask  = Qmask(bp="*!", k=2)
3 mask  = Qmask(bp="!*", k=2)
4 mask  = Qmask(bp="!*!",k=2)
5 mask  = Qmask(bp="01", k=2)
6 Qinit(8) + (cycle + mask)*3
```



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Compressed *Algorithmic* Descriptions

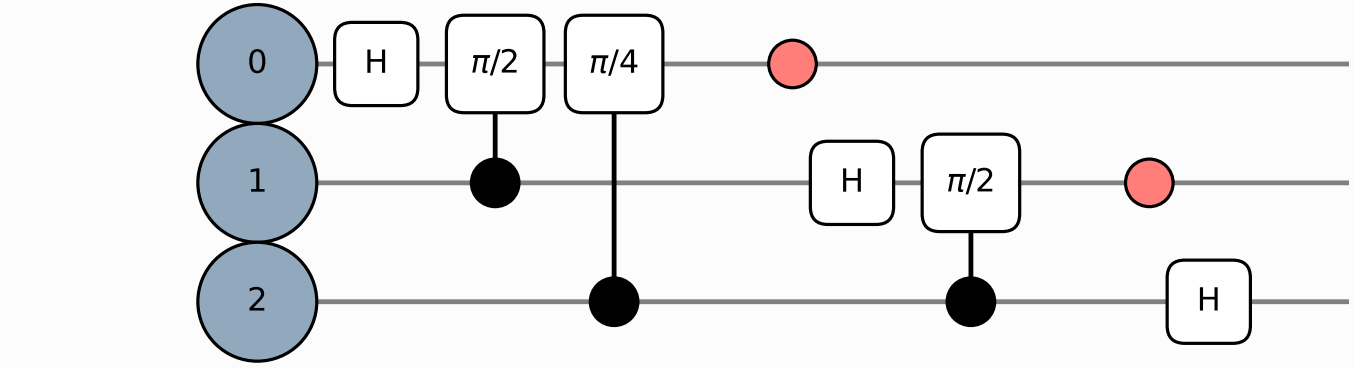
```
1 cycle = Qcycle(step=1, stride=1, k=2, boundary=0)
2 mask  = Qmask(bp="*!", k=2)
3 mask  = Qmask(bp="!*", k=2)
4 mask  = Qmask(bp="!*!",k=2)
5 mask  = Qmask(bp="01", k=2)
6 Qinit(8) + (cycle + mask)*3
```

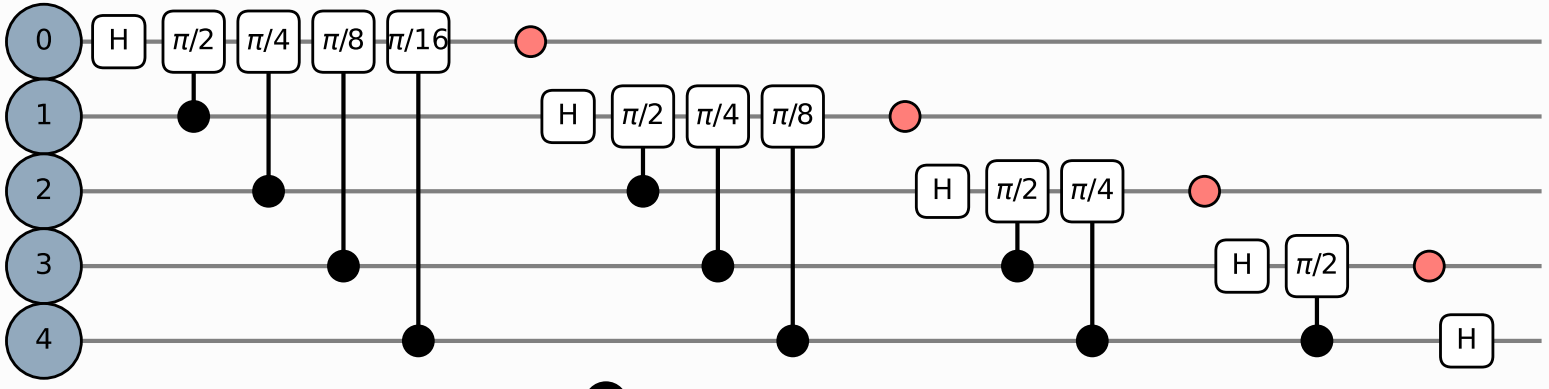


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HierarQcal Example

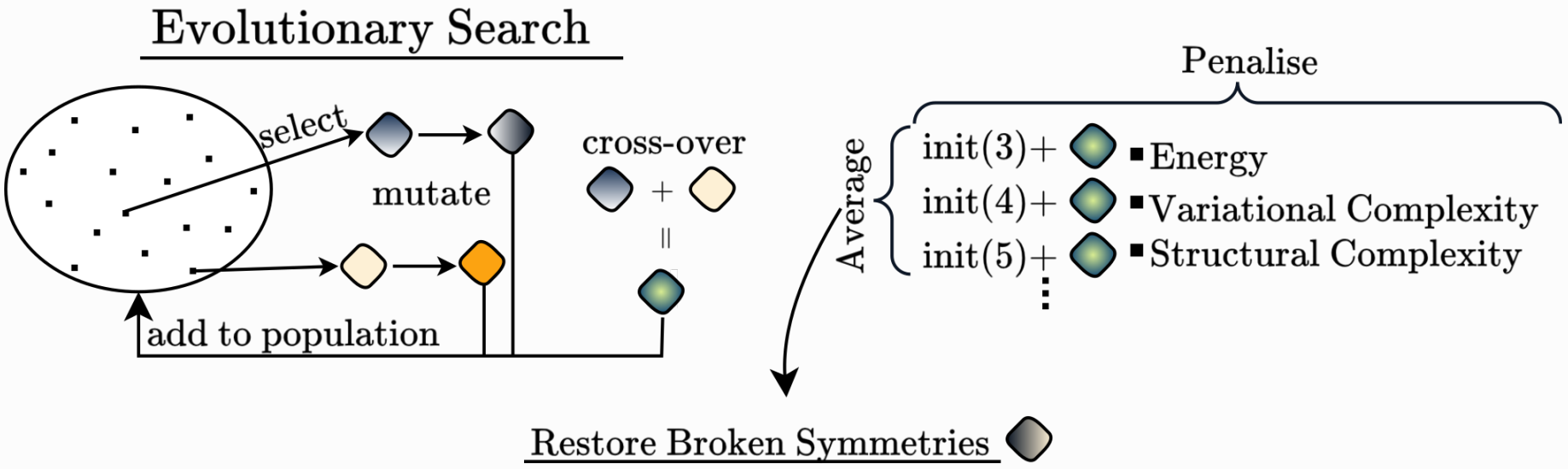
```
1 htop = Qpivot(bp = "1*", map=H)
2 pivot_phase = Qpivot(bp = "1*", map=CP(ϕ), symbol_fn=π*2^(-ne))
3 mask_top = Qmask(bp = "1*")
4 qft = (htop + pivot_phase + mask_top)*N
5 #=================================================#
6 Qinit(3) + qft                                Qinit(5) + qft
```

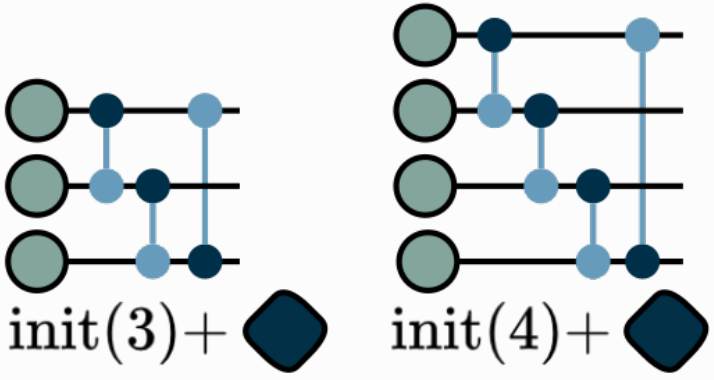




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Generating Generalised Ground-State Ansätze





Primitive Operations: $e^{-i\theta A \otimes B}$ where $A, B \in \{I, \frac{1}{2}X, \frac{1}{2}Y, \frac{1}{2}Z\}$

M. Lourens, I. Sinayskiy, J. N. Kriel, and F. Petruccione, *Phys. Rev. Lett.* **135**, 210401 (2025).

Generating Generalised Ground-State Ansätze

Lipkin-Meshkov-Glick (LMG) Hamiltonian:

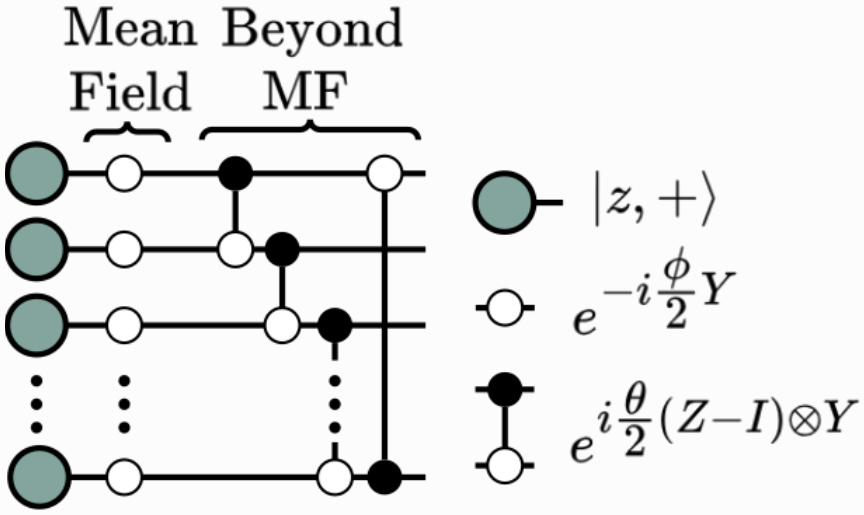
$$H_{\mathrm{LMG}} = -\frac{J}{4N}\sum_{i<j} Z_i Z_j - \frac{h}{2}\sum_{i=0}^{N-1} X_i, \tag{1}$$

Transverse-Field Ising Model (TFIM) Hamiltonian:

$$H_{\mathrm{TFIM}} = -\frac{J}{4}\sum_{i=0}^{N-1} Z_i Z_{i+1} - \frac{h}{2}\sum_{i=0}^{N-1} X_i. \tag{2}$$

M. Lourens, I. Sinayskiy, J. N. Kriel, and F. Petruccione, *Phys. Rev. Lett.* **135**, 210401 (2025).

Generating Generalised Ground-State Ansätze

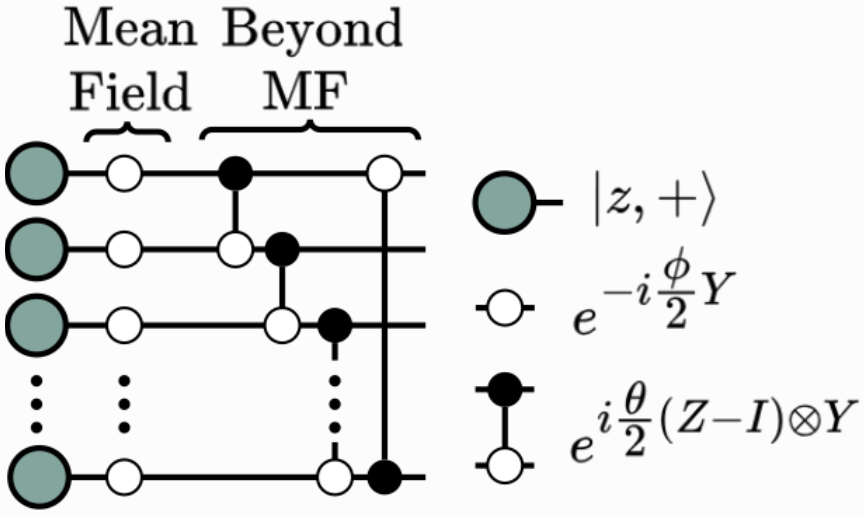


$$|\psi_o\rangle = 2^{-N/2} \sum_{\vec{\sigma}} \mathrm{Tr}\left(B^{\sigma_0} A^{\sigma_1} \cdots A^{\sigma_{N-1}}\right) |y, \vec{\sigma}\rangle \,, \tag{3}$$

$$A^{+} = \begin{bmatrix} \cos\frac{\theta}{2} & \cos\frac{\theta}{2} \\ i\sin\frac{\theta}{2} & -i\sin\frac{\theta}{2} \end{bmatrix} \begin{bmatrix} e^{-\frac{i(\theta+\phi)}{2}} & 0 \\ 0 & e^{\frac{i(\theta+\phi)}{2}} \end{bmatrix}, A^{-} = (A^{+})^{*}. \tag{4}$$

M. Lourens, I. Sinayskiy, J. N. Kriel, and F. Petruccione, *Phys. Rev. Lett.* **135**, 210401 (2025).

Generating Generalised Ground-State Ansätze



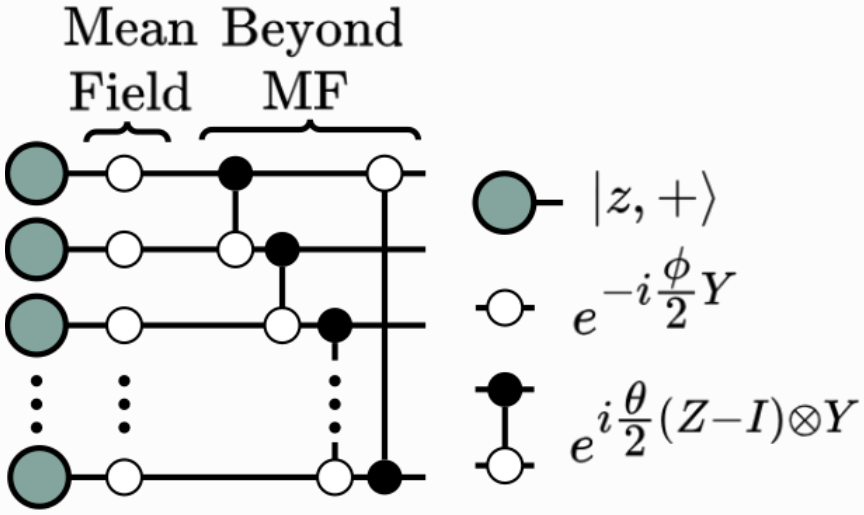
Not translationally invariant

$$|\psi_o\rangle = 2^{-N/2} \sum_{\vec{\sigma}} \mathrm{Tr}(B^{\sigma_0} A^{\sigma_1} \cdots A^{\sigma_{N-1}}) \, |y, \vec{\sigma}\rangle \, , \tag{3}$$

$$A^+ = \begin{bmatrix} \cos\frac{\theta}{2} & \cos\frac{\theta}{2} \\ i\sin\frac{\theta}{2} & -i\sin\frac{\theta}{2} \end{bmatrix} \begin{bmatrix} e^{-\frac{i(\theta+\phi)}{2}} & 0 \\ 0 & e^{\frac{i(\theta+\phi)}{2}} \end{bmatrix}, A^- = (A^+)^*. \tag{4}$$

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Generating Generalised Ground-State Ansätze



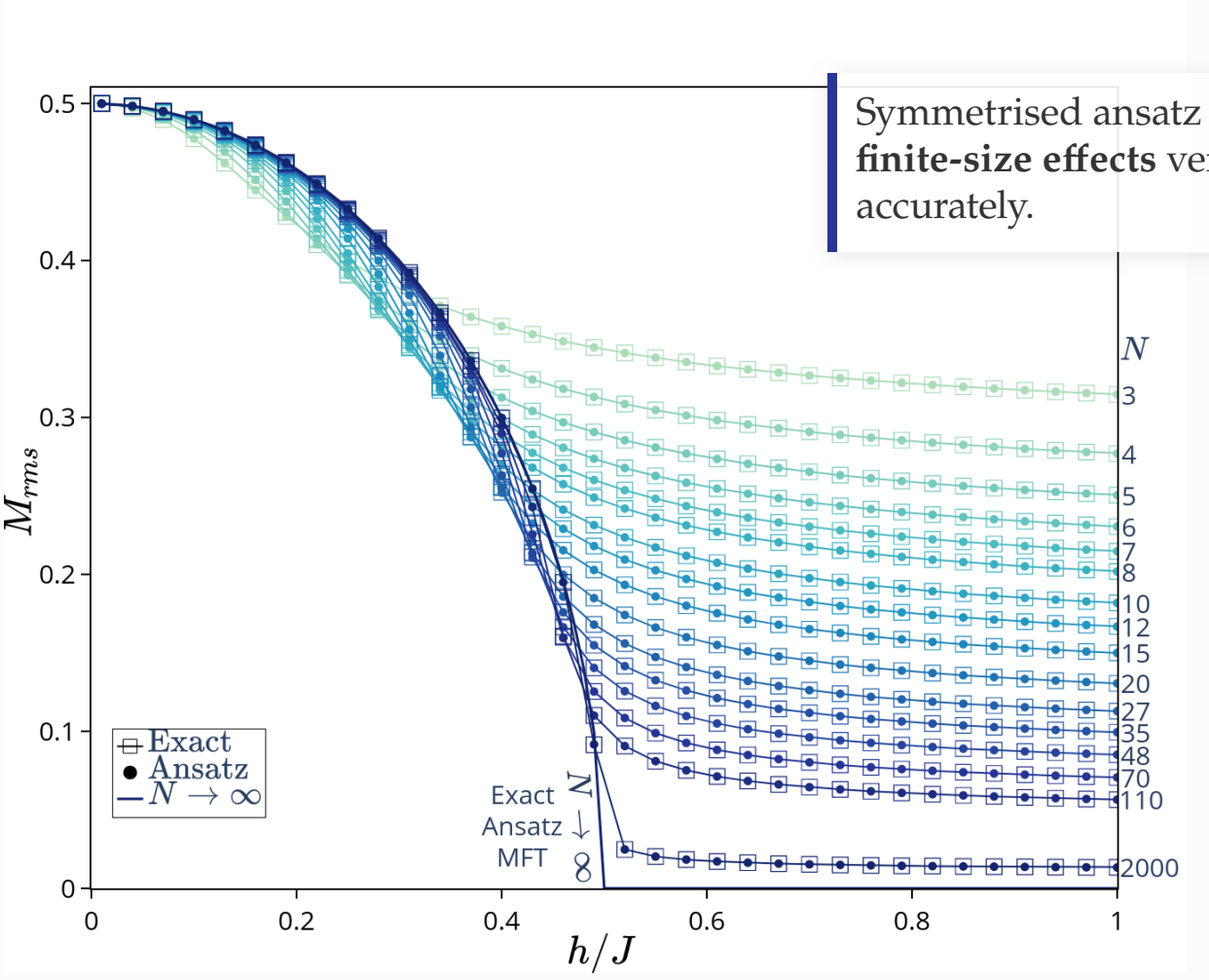
Not translationally invariant

$$|\psi_o\rangle = 2^{-N/2} \sum_{\vec{\sigma}} \mathrm{Tr}\left(B^{\sigma_0} A^{\sigma_1} \cdots A^{\sigma_{N-1}}\right) |y, \vec{\sigma}\rangle \,, \tag{3}$$

$$|\psi_t\rangle = \frac{1}{M} \sum_{\vec{\sigma}} \mathrm{Tr}\left(A^{\sigma_0} \cdots A^{\sigma_{N-1}}\right) |y, \vec{\sigma}\rangle \,. \tag{5}$$

M. Lourens, I. Sinayskiy, J. N. Kriel, and F. Petruccione, *Phys. Rev. Lett.* **135**, 210401 (2025).

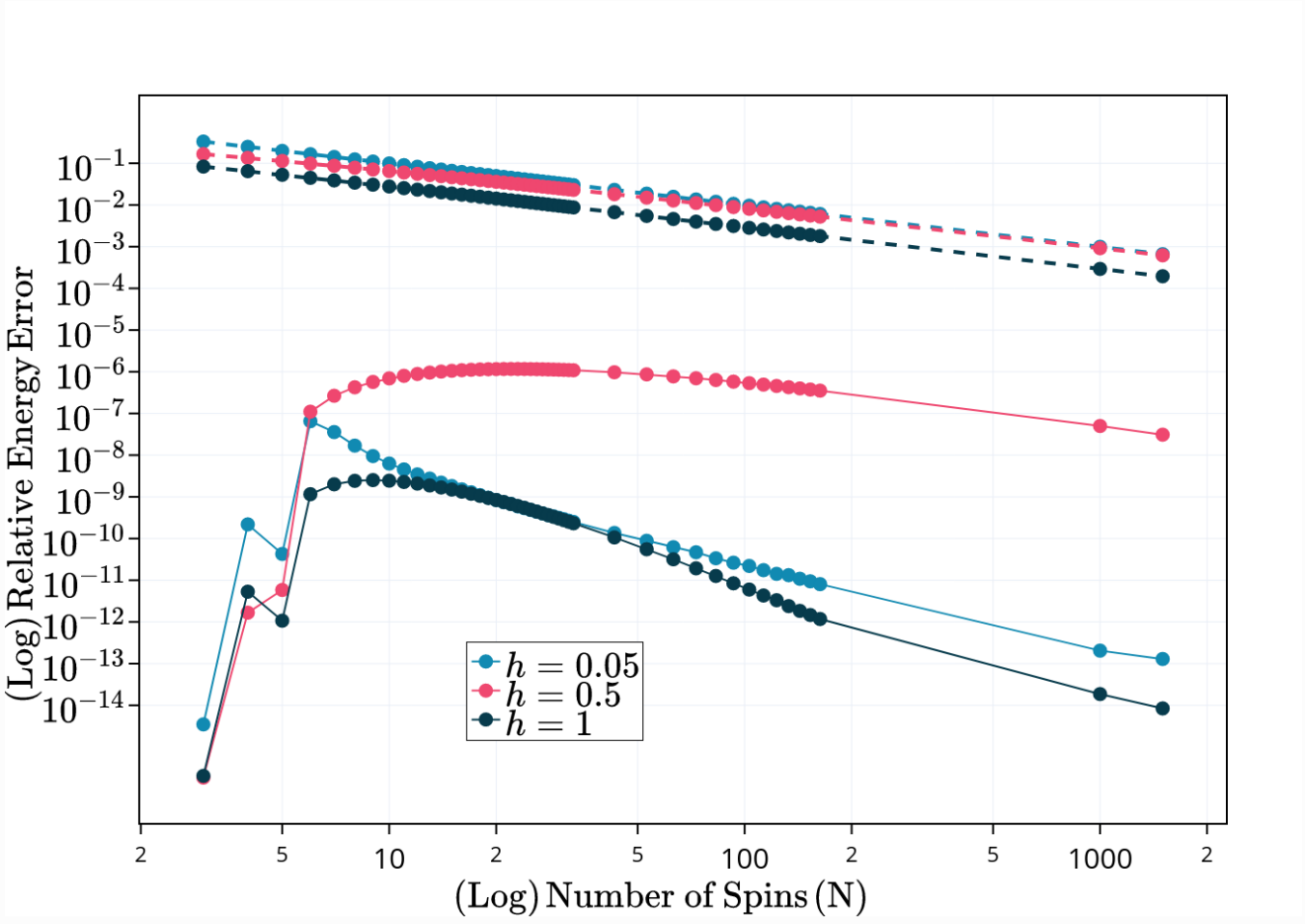
LMG Results - *RMS Magnetisation*



$$M_{\text{rms}} = \frac{1}{2N}\sqrt{\left\langle \left(\sum_i Z_i\right)^2 \right\rangle}. \tag{6}$$

M. Lourens, I. Sinayskiy, J. N. Kriel, and F. Petruccione, *Phys. Rev. Lett.* **135**, 210401 (2025).

LMG Results - *Relative Energy Error*

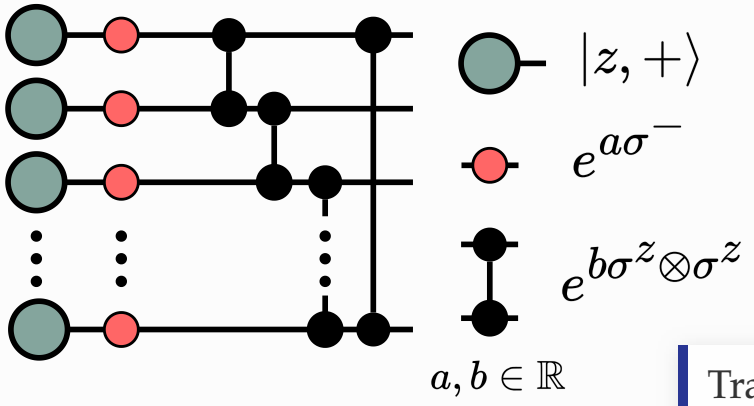


$$\epsilon_{\text{rel}} = \frac{|E_{\text{pred}} - E_{\text{exact}}|}{|E_{\text{exact}}|} \tag{7}$$

M. Lourens, I. Sinayskiy, J. N. Kriel, and F. Petruccione, *Phys. Rev. Lett.* **135**, 210401 (2025).

Different *Operator Basis*

Primitive Operations: $e^{\theta A \otimes B}$ where $A, B \in \{I, \sigma^+, \sigma^-, Z\}$



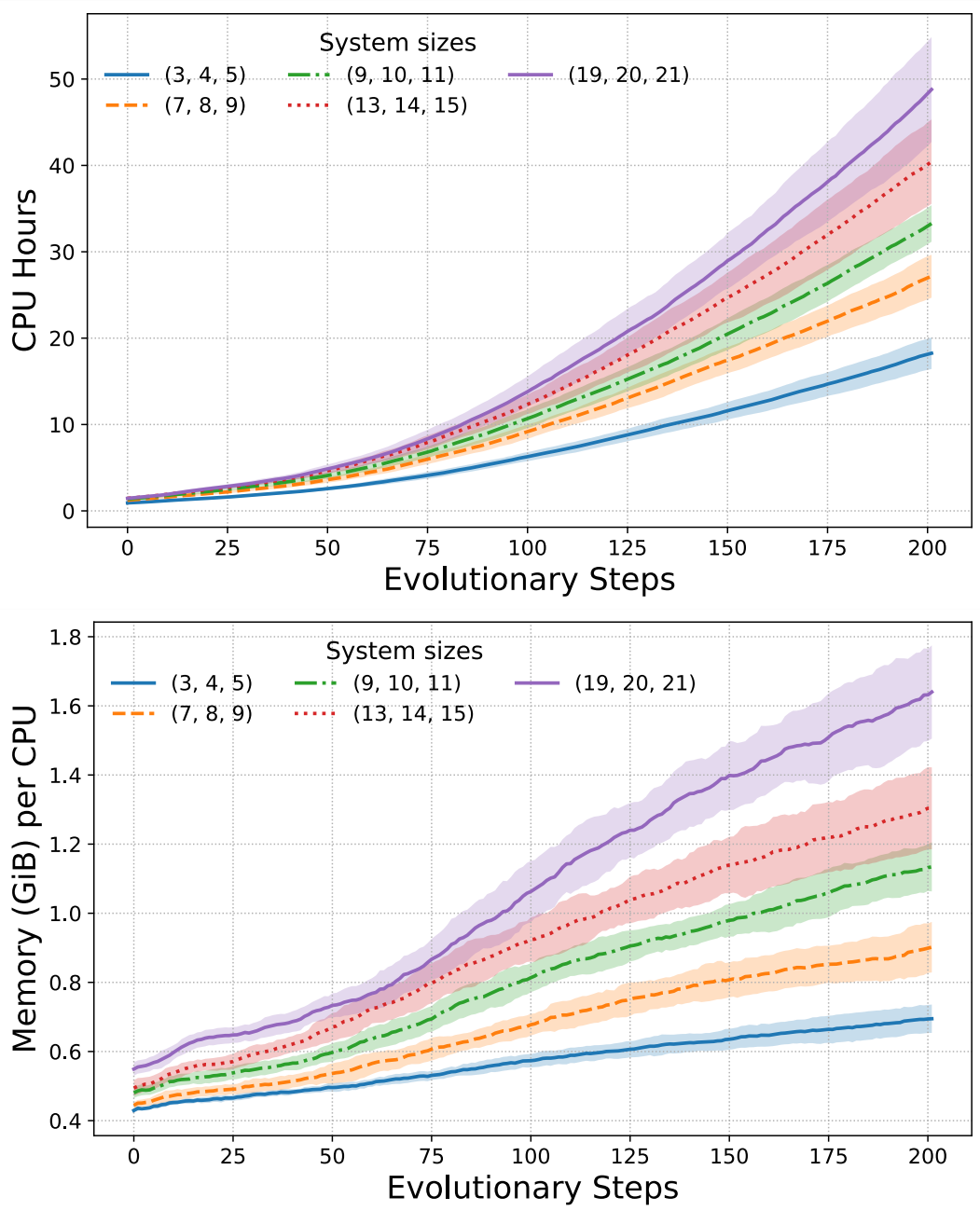
Translationally invariant ansatz

$$|\psi_l\rangle = \frac{1}{M_l} \sum_{\vec{\sigma}} \mathrm{Tr}\left(C^{\sigma_0} C^{\sigma_1} \cdots C^{\sigma_{N-1}}\right) |z, \vec{\sigma}\rangle \tag{8}$$

$$C^{\pm} = a^{\frac{1 \mp 1}{2}} \begin{bmatrix} \cosh(b) & \pm\cosh(b) \\ \pm\sinh(b) & \sinh(b) \end{bmatrix} \tag{9}$$

M. Lourens, I. Sinayskiy, J. N. Kriel, and F. Petruccione, *Phys. Rev. Lett.* **135**, 210401 (2025).

Search Strategy - *Computational Cost*



M. Lourens, I. Sinayskiy, J. N. Kriel, and F. Petruccione, *Phys. Rev. Lett.* **135**, 210401 (2025).

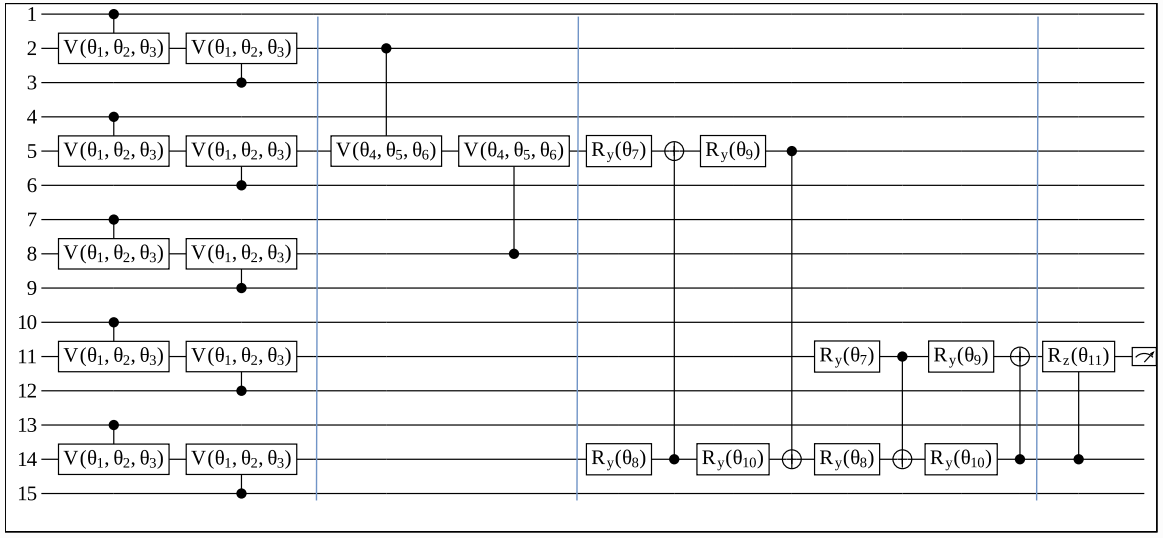
Quantum Phase Recognition

$$H = -J \sum_{i=1}^{N-2} Z_i X_{i+1} Z_{i+2} - h_1 \sum_{i=1}^{N} X_i - h_2 \sum_{i=1}^{N-1} X_i X_{i+1}.$$

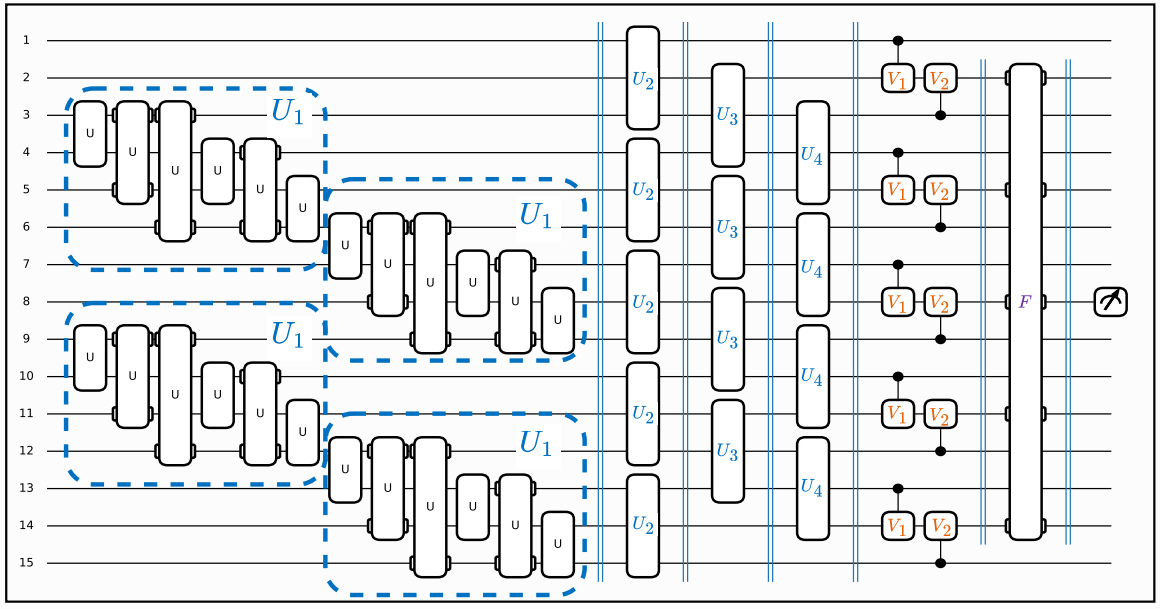
- Possible phases for a ground state $|\psi_g\rangle$:
  - Paramagnetic
  - $\mathbb{Z}_2 \times \mathbb{Z}_2$ Symmetry-Protected Topological (SPT)
  - Antiferromagnetic
- Given an unknown ground state $|\psi_g\rangle$, is it in a SPT phase?

Quantum Phase Recognition

M. Lourens, I. Sinayskiy, D. K. Park, C. Blank, and F. Petruccione, *npj Quantum Inf.* **9**, 79 (2023).

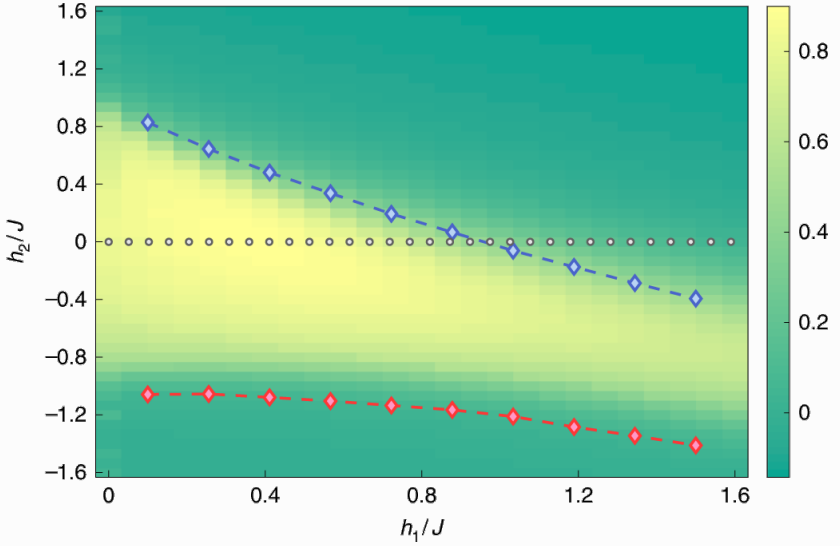


I. Cong, S. Choi, and M. D. Lukin, *Nat. Phys.* **15**, 1273 (2019).

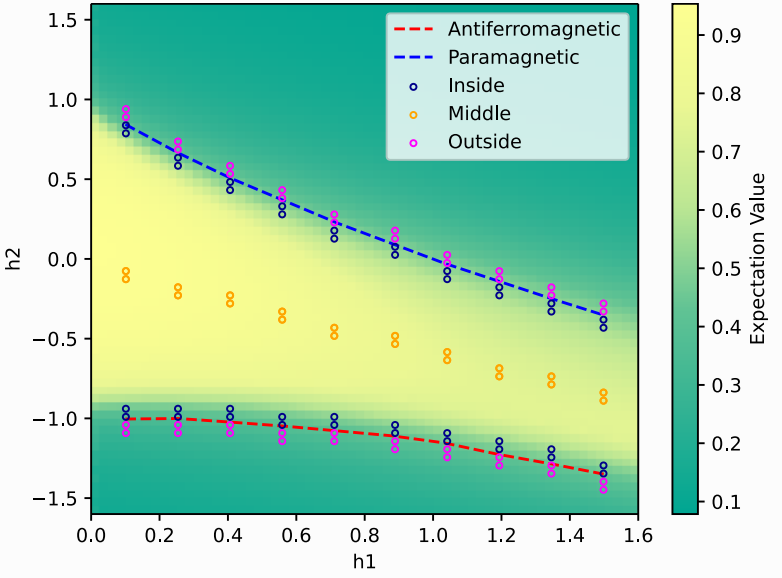


Quantum Phase Recognition

Reference[1]



Found[2]



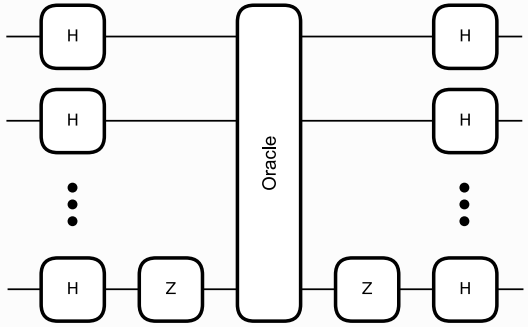
Metric	Reference	Found
Number of parameters	1308	**11**
Sample Complexity (Inside)	61.523	**36.079**
Sample Complexity (Middle)	**10.992**	13.253
MSE (Outside)	**0.164**	0.167

[1] I. Cong, S. Choi, and M. D. Lukin, *Nat. Phys.* **15**, 1273 (2019).

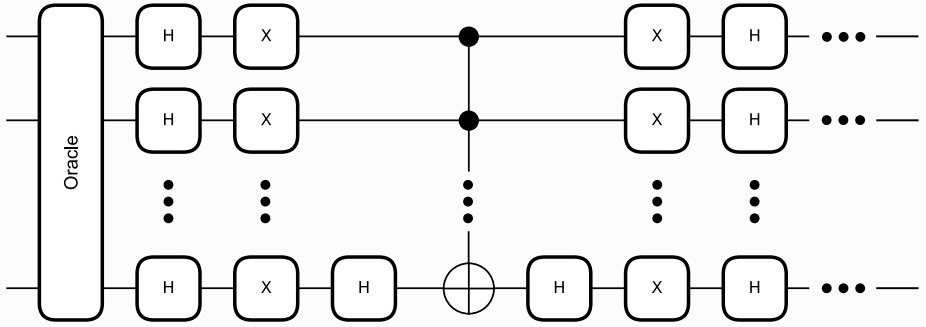
[2] **M. Lourens**, I. Sinayskiy, D. K. Park, C. Blank, and F. Petruccione, *npj Quantum Inf.* **9**, 79 (2023).

Quantum Algorithm Synthesis

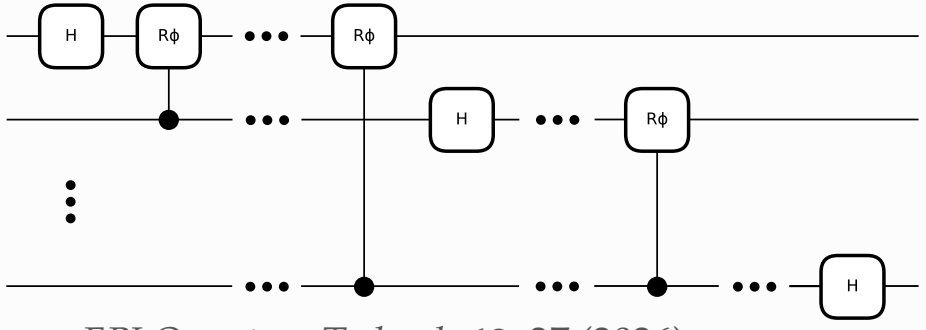
```
dj= (
    Qcycle(map=H)
  + Qpivot(bp="*1" ,map=Z)
  + Oracle
  + Qpivot(bp="*1" ,map=Z)
  + Qcycle(map=H)
)*r
```




```
grover = (
  Oracle
  + Qcycle(map=H)
  + Qcycle(map=X)
  + Qpivot(bp="*1", map=H)
  + Qpivot(bp="*1", map=MCX)
  + Qpivot(bp="*1", map=H)
  + Qcycle(map=X)
  + Qcycle(map=H)
)*r
```

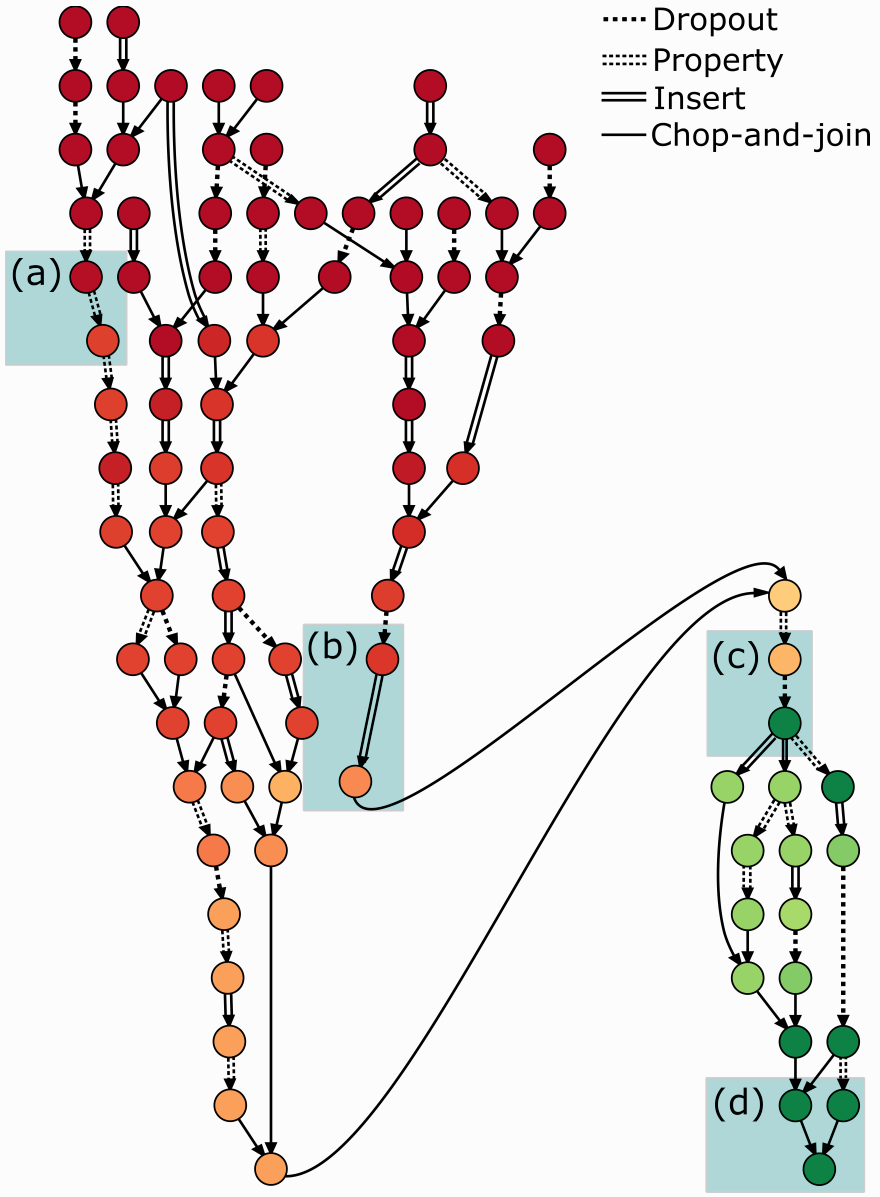



```
qft = (
    Qpivot(bp="1*", map=H)
  + Qpivot(bp="1*", map=CP(ϕ))
  + Qmask(bp="1*")
)*r
```



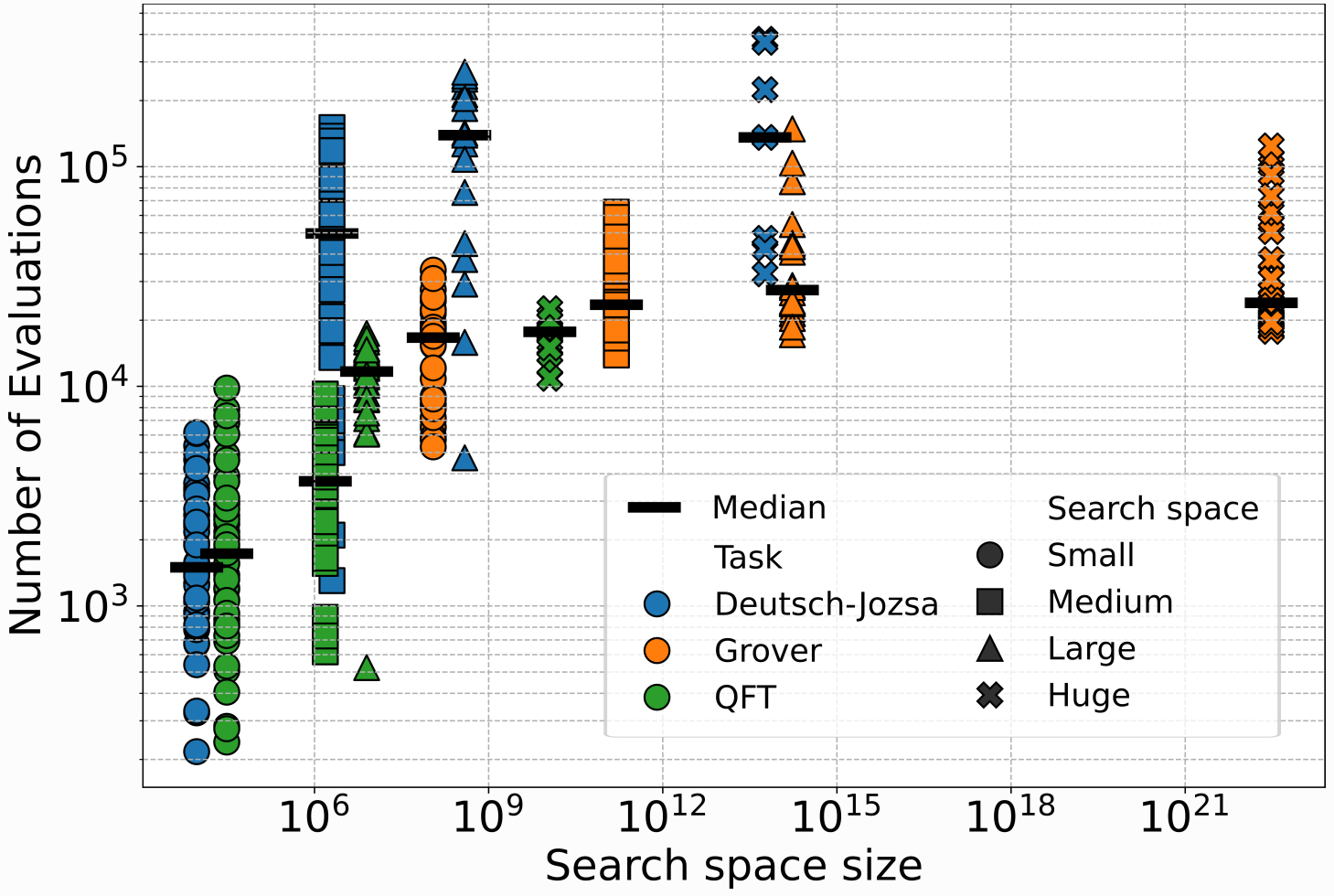
A. Rouillard, **M. Lourens**, and F. Petruccione, *EPJ Quantum Technol.* **13**, 27 (2026).

Quantum Algorithm Synthesis



A. Rouillard, **M. Lourens**, and F. Petruccione, *EPJ Quantum Technol.* **13**, 27 (2026).

Quantum Algorithm Synthesis



A. Rouillard, **M. Lourens**, and F. Petruccione, *EPJ Quantum Technol.* **13**, 27 (2026).

Summary

Key Points

- DSL to build tensor networks
  - Algorithmic description
  - Capture system size scaling
  - Modular composition
- Evolutionary search
  - Penalise complexity
  - Reward accuracy
- Symmetry Restoration

Papers

M. Lourens, I. Sinayskiy, D. K. Park, C. Blank, and F. Petruccione, "Hierarchical quantum circuit representations for neural architecture search," *npj Quantum Inf.* **9**, 79 (2023).

M. Lourens, I. Sinayskiy, J. N. Kriel, and F. Petruccione, "Generating Generalized Ground-State Ansätze from Few-Body Examples," *Phys. Rev. Lett.* **135**, 210401 (2025).

A. Rouillard, **M. Lourens**, and F. Petruccione, "Automated quantum algorithm design using a domain-specific language," *EPJ Quantum Technol.* **13**, 27 (2026).

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